

Modeling a MIP Problem: Optimal Placement of Shared Umbrella stands on Campus

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Part I: Introduction

Background

Suzhou has a warm and humid climate with abundant rainfall and unpredictable weather. Teachers and students in XJTLU often face the challenge of "going out on sunny days and encountering a sudden rain", which creates many inconveniences in their daily lives. Some students may have noticed that the central building (CB) offers around 50 umbrellas for rental. However, this existing umbrella rental service is insufficient due to its limited quantity and the singular borrowing and returning locations, which do not satisfy the urgent needs of teachers and students during unexpected rain. During peak rainfall, many teachers and students have no choice but to wait in the rain or move forward in the rain. This situation not only affects normal learning and living order, but may also lead to teachers and students catching colds and getting sick due to exposure to rain.

To effectively address the inconvenience caused by sudden rainfall, improve the campus experience, and improve the quality of campus service, it is essential to establish convenient umbrella rental stands near different building areas in XJTLU. By rationally planning the stands of rental stands, such as setting up rental stands around places with high foot traffic, like teaching buildings and libraries, and providing a sufficient number of umbrellas, we can better meet the needs of both teachers and students during unexpected rain showers.

Part II: Problem Description

Problem Raised

The layout planning for the umbrella rental stands must carefully consider various factors, such as the costs of purchasing the umbrellas and the time it takes for teachers and students to travel to the rental stands. In this situation, we pose an important question: How can we effectively locate and establish umbrella rental stands on campus to minimize the overall construction costs?

Influencing Factors

Firstly, we identify the teaching buildings in XJTLU that require umbrellas on rainy days as the "potential demand stands" in our model. The demand for umbrellas at each of these stands is estimated based on the frequency of usage of the buildings and the flow of people passing through them.

Additionally, we have selected several potential stands for umbrella rental stands, referred to as candidate stands. These stands are strategically distributed in areas with convenient transportation on campus. The establishment of each umbrella rental stand incurs certain costs, which include expenses for site hardware and the umbrellas themselves.

It is important to note that the capacity of each stand is limited and cannot accommodate unlimited user demands. We have also recorded the walking time between each demand stands and each candidate rental stand to evaluate user convenience. The time it takes for teachers and students to walk to the umbrella rental stand is closely related to the stand of that stand. Therefore, rental stands should be situated in areas with high foot traffic whenever possible to minimize walking time for users.

Our estimation of time costs must consider the geographic locations of each building on campus as well as crowd flow. Lastly, given practical constraints like budget and site availability, the school can only construct a limited number of umbrella rental stands.

Goal Setting

Based on the above analysis, this paper aims to establish a mixed-integer planning model to optimize the location and size of umbrella rental stands to achieve the combined minimization of construction and time costs, so as to enhance the efficiency and convenience of campus umbrella rental services and provide better rainy day travel protection for teachers and students. Specifically, the model will take into account various factors such as construction costs (procurement costs of umbrellas and umbrella stands) and time costs to achieve an optimal layout of the campus umbrella rental service.

Part III: Construction of the MIP Optimization Model

We set up several sets S, B, U to represent the buildings, potential umbrella rental stands, and the umbrella capacity in XJTLU.

- Let S be the set of buildings in the XJTLU campus where students and teachers stay on rainy days. The buildings include: FB, CB, SA, SB, SC, SD, PB, MA, MB, EB, EE, AS, IR, BS, IA, HS, ES, DB, GYM.
- Let B be the set of potential locations for each umbrella rental stand. The potential locations are marked by the red numbers 1-19 in the figure 1. There are corridors that connect different buildings, so you can go to the umbrella rental stand without getting wet in the rain.
- Let U be the set of umbrella capacity provided by each umbrella rental stand.



Figure 1: Selection of potential locations for umbrella rental stands

We establish variable notations in order to prepare for the construction of the objective function that follows.

- Let d_i denote the total demand of students and teachers i , where $i \in S$.
- Let c denote the cost of each umbrella set (an umbrella set includes the umbrella and the standardized umbrella rental stand).
- Let n denote the maximum number of umbrella rental stands that can be set in XJTLU.

- Let m_j denote the umbrella capacity at location j
- Let u_j denote the each umbrella rental stand capacity j , where $j \in B$.
- Let t_{ij} denote the transportation time for students and teachers who need to borrow an umbrella from the building i to the location j , where $i \in S$ and $j \in B$.
- Let x_{ij} be defined as

$$x_{ij} = \begin{cases} 1, & \text{if the demand of students and teachers from building } i \text{ is satisfied by potential location } j; \\ 0, & \text{otherwise.} \end{cases}$$

where $i \in S$ and $j \in B$.

- Let y_j be defined as

$$y_j = \begin{cases} 1, & \text{if an umbrella rental stand is build at location } j; \\ 0, & \text{otherwise.} \end{cases}$$

To simplify the modeling process and to ensure the feasibility of the optimized model, we make the following basic assumptions in our analysis:

- The cost of students to go to the rental stand is directly proportional to the time.
- The time for students to go to the rental stand is proportional to the distance.
- All umbrella demand originates from the set S .
- Umbrellas can only be stored at candidate stands B .
- Construction cost is directly proportional to the purchase price of the umbrella set.
- The number of installed stands $\leq n$.
- Umbrellas at stand $j \leq m_j$.
- The student and teacher from each building are assigned to exactly one stand.
- Installation cost and User Inconvenience Cost are unitless for direct optimization.

The objective of this model is to minimize the total system cost, which consists of the user inconvenience cost and the umbrella station construction cost, as shown below.

$$\text{Minimize} \quad \sum_{i \in S} \sum_{j \in B} t_{ij} d_i x_{ij} + \sum_{j \in B} c u_j \quad (1)$$

To ensure the feasibility and logical consistency of the model, the following constraints need to be met

- (1) Students from the same building i must be assigned to a single rental stand j :

$$\sum_{j \in B} x_{ij} = 1 \quad \forall i \in S \quad (2)$$

- (2) At most n rental stands can be selected from predefined locations:

$$\sum_{j \in B} y_j \leq n \quad (3)$$

- (3) Total demand at rental stand j must not exceed its capacity:

$$\sum_{i \in S} d_i x_{ij} \leq u_j y_j \quad \forall j \in B \quad (4)$$

- (4) Capacity of each rental stand j must not exceed the maximum m :

$$u_j \leq m y_j \quad \forall j \in B \quad (5)$$

Part IV: Data Analysis

We use Excel to store hypothetical data regarding the demand for umbrellas, factoring in time coefficients. Matlab was utilized for conducting a sensitivity analysis of the data, and the built-in intlinprog solver was employed to obtain the model's optimization results.

Demand Simulation Settings

This study simulates the demand for umbrellas in each teaching building (including the North and South Campuses) based on the following assumptions: the average daily flow of students and teachers is estimated based on the floor area of each teaching building, and it is assumed that there is a positive correlation between the two, i.e., the larger the area of the teaching building, the higher the average daily flow of students and teachers.

$$\text{Foot traffic} \propto \text{Building area}$$

The average daily demand for umbrellas in each building is then estimated based on the average daily flow of students and teachers. Under the premise that weather changes will cause changes in the demand for umbrellas, the demand for each building is dynamically adjusted, with appropriate upward fluctuations on rainy days and corresponding downward adjustments on sunny days. The total demand for umbrellas remained relatively stable over the six-day period, with a daily average of about 547 umbrellas, and fluctuations were set within a reasonable range.

di	Day1	Day2	Day3	Day4	Day5	Day6
FB	40	50	45	60	80	30
CB	60	55	35	45	50	60
SA	15	20	20	10	15	25
SB	20	25	30	15	20	25
SC	15	20	10	20	5	15
SD	20	15	18	17	16	19
PB	20	25	22	23	34	21
MA	30	32	26	50	20	15
MB	30	28	44	52	38	22
EB	25	43	21	34	26	27
EE	20	34	21	42	28	36
AS	15	11	33	23	16	17
IR	10	14	15	12	14	13
IA	30	22	33	24	31	27
BS	70	66	58	62	73	68
HS	15	14	22	17	12	24
ES	30	26	27	28	29	31
DB	30	22	35	28	34	21
GYM	20	24	32	31	25	17
	515	546	547	593	566	513

Figure 2: Demand Simulation Settings

Temporal Parameter Configuration

It is assumed that the time to travel from each building to each umbrella rental stand is proportional to the distance between them. The travel times from the FB building to rental stand 17 (the furthest distance) and the ES building to rental stand 1 (the furthest distance) were set as a baseline value of 1 time unit. The passage time from other academic buildings to each leasing stand is calculated proportionally according to the relative distance, so as to simulate the time for other academic buildings to arrive at each umbrella leasing stand, and to ensure that the time parameter maintains a positive relationship with the spatial distance. This can be expressed as using the formula:

$$t_{ij} \propto d_{ij} \quad \forall i \in \text{Buildings}, j \in \text{stands}$$

$$t_{ij} = \frac{d_{ij}}{d_{\text{base}}} \times t_{\text{base}}$$

where d_{base} represents the maximum reference distance.

ij	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
FB	0.1	0.3	0.5	0.4	0.35	0.45	0.5	0.55	0.6	0.6	0.85	0.75	0.9	0.8	1	0.95	0.9		
CB	0.25	0.1	0.3	0.25	0.25	0.4	0.45	0.5	0.5	0.4	0.65	0.55	0.7	0.55	0.7	0.6	0.8	0.75	0.7
SA	0.3	0.25	0.2	0.1	0.05	0.2	0.25	0.3	0.4	0.35	0.6	0.5	0.65	0.5	0.65	0.55	0.75	0.7	0.65
SB	0.35	0.2	0.2	0.05	0.05	0.15	0.2	0.25	0.3	0.25	0.5	0.4	0.55	0.4	0.55	0.45	0.65	0.6	0.55
SC	0.4	0.25	0.1	0.1	0.15	0.2	0.25	0.3	0.25	0.15	0.4	0.3	0.45	0.3	0.45	0.35	0.55	0.5	0.45
SD	0.5	0.3	0.05	0.2	0.25	0.3	0.35	0.4	0.3	0.15	0.4	0.3	0.45	0.3	0.45	0.35	0.55	0.5	0.45
PB	0.4	0.35	0.3	0.2	0.15	0.05	0.1	0.15	0.3	0.25	0.5	0.4	0.55	0.4	0.55	0.45	0.65	0.6	0.55
MA	0.45	0.45	0.35	0.25	0.25	0.15	0	0.05	0.2	0.25	0.5	0.4	0.55	0.4	0.55	0.45	0.65	0.6	0.55
MB	0.55	0.5	0.4	0.3	0.3	0.2	0.15	0.1	0.15	0.25	0.5	0.4	0.55	0.4	0.55	0.45	0.65	0.6	0.55
EB	0.6	0.5	0.3	0.3	0.35	0.25	0.2	0.15	0	0.15	0.4	0.3	0.45	0.3	0.45	0.35	0.55	0.5	0.45
EE	0.6	0.4	0.15	0.2	0.25	0.2	0.25	0.25	0.15	0.05	0.3	0.2	0.35	0.2	0.35	0.25	0.45	0.4	0.35
AS	0.85	0.65	0.4	0.5	0.55	0.5	0.55	0.55	0.4	0.3	0	0.1	0.2	0.25	0.4	0.2	0.4	0.45	0.5
IR	0.7	0.5	0.3	0.35	0.4	0.35	0.4	0.4	0.3	0.2	0.1	0	0.15	0.15	0.3	0.15	0.4	0.4	0.4
IA	0.7	0.5	0.3	0.35	0.4	0.35	0.4	0.4	0.3	0.2	0.25	0.15	0.25	0.05	0.2	0.1	0.3	0.25	0.25
BS	0.9	0.7	0.5	0.55	0.6	0.55	0.6	0.6	0.5	0.4	0.2	0.2	0.05	0.25	0.2	0.1	0.15	0.2	0.4
HS	0.85	0.65	0.45	0.5	0.55	0.5	0.55	0.55	0.45	0.35	0.4	0.3	0.2	0.1	0.15	0.1	0.15	0.1	0.3
ES	1	0.8	0.6	0.65	0.7	0.65	0.7	0.7	0.6	0.5	0.4	0.35	0.2	0.35	0.2	0.25	0.1	0.2	0.5
DB	0.95	0.75	0.55	0.6	0.65	0.6	0.65	0.65	0.55	0.45	0.45	0.35	0.3	0.25	0.15	0.25	0.2	0.1	0.4
GYM	0.85	0.65	0.45	0.5	0.55	0.5	0.55	0.55	0.45	0.35	0.5	0.4	0.45	0.2	0.25	0.3	0.5	0.4	0.05

Figure 3: Temporal Parameter Configuration

Table 1: Optimization Results

Day	Total Cost	Enabled Rental stands	Capacity Configuration
Day1	12919.25	1 2 3 5 7 9 11 13 14 18	stand 1: 40, stand 2: 60, stand 3: 35, stand 5: 35, stand 7: 50, stand 9: 75, stand 11: 25, stand 13: 70, stand 14: 50, stand 18: 75
Day2	13694.10	1 2 5 7 9 10 12 13 18 19	stand 1: 50, stand 2: 55, stand 5: 45, stand 7: 57, stand 9: 71, stand 10: 69, stand 12: 47, stand 13: 66, stand 18: 62, stand 19: 24
Day3	13721.45	1 2 3 5 8 11 13 14 18 19	stand 1: 45, stand 2: 35, stand 3: 49, stand 5: 72, stand 8: 91, stand 11: 48, stand 13: 58, stand 14: 33, stand 18: 84, stand 19: 32
Day4	14874.65	1 2 4 7 9 10 12 13 18 19	stand 1: 60, stand 2: 45, stand 4: 45, stand 7: 73, stand 9: 86, stand 10: 59, stand 12: 59, stand 13: 62, stand 18: 73, stand 19: 31
Day5	14199.95	1 2 5 7 9 10 12 13 18 19	stand 1: 80, stand 2: 50, stand 5: 35, stand 7: 92, stand 9: 26, stand 10: 49, stand 12: 61, stand 13: 73, stand 18: 75, stand 19: 25
Day6	12867.95	1 2 5 7 9 10 11 13 14 17	stand 1: 30, stand 2: 60, stand 5: 65, stand 7: 58, stand 9: 27, stand 10: 55, stand 11: 30, stand 13: 68, stand 14: 68, stand 17: 52

Sensitivity analysis

Figure 4 illustrates the sensitivity analysis of the installed cost c to the average total daily cost of the system. It is evident from the figure that the total cost of the system tends to increase linearly as the unit construction cost increases from 20 to 30, with an overall increase of over 5,000 units. This indicates that the model is highly sensitive to changes in construction cost. For actual deployment, cost control should be given high priority to avoid unnecessary budget inflation. In addition, the linear response characteristic of this cost term provides a convenient prediction basis for the evaluation of the project budget.

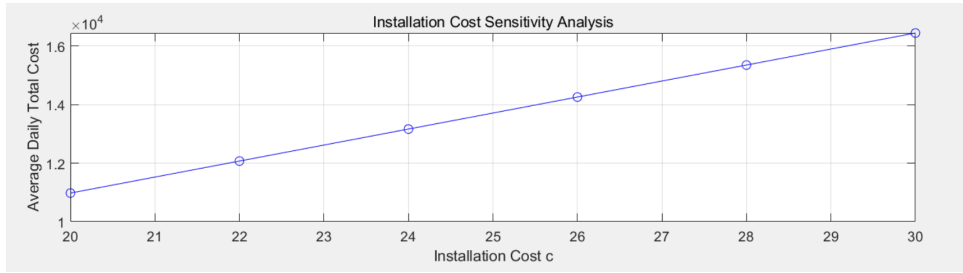


Figure 4: Installation Cost Sensitivity Analysis

Figure 5 shows the sensitivity analysis of the maximum number of umbrella rental standss (n) on the average total daily cost.

As n increases, the total cost decreases due to enhanced flexibility in stand placement, which reduces user inconvenience without changing demand. When n is between 6 and 10, each additional rental stand significantly improves service coverage and shortens user walking time, leading to a notable cost reduction. However, when n increases from 10 to 15, the cost continues to decrease at a slower rate, indicating a reasonable system layout where adding more stands offers limited optimization.

Notably, with an initial setting of $n = 5$, capacity was insufficient to meet demand, making the model unsolvable, which highlights the importance of having at least six rental stands to ensure system feasibility.

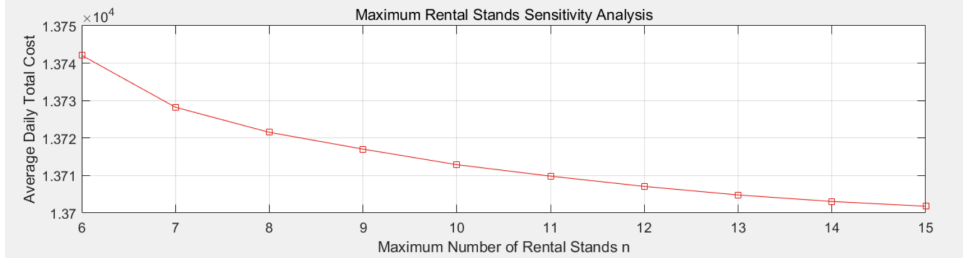


Figure 5: Maximum Rental stands Sensitivity Analysis

Figure 6 illustrates the “Capacity per stand Sensitivity Analysis”. Let m denote the capacity per stand and C_{avg} denote the average daily total cost. Overall, it shows a trend of first decreasing and then stabilizing.

When $m = 80$, $C_{avg} \approx 1.37133 \times 10^4$. As m increases from 80 to 90, C_{avg} decreases significantly, reflecting that increasing the capacity of each rental stand at this stage can effectively reduce the average daily total cost. Since the capacity of some of our umbrellas $m > 80$, the fluctuation of m within the interval $[80, 90]$ will largely affect the inconvenience cost for demanders in XJTLU. When $m \geq 90$, $C_{avg} \approx 1.37129 \times 10^4$. Even as m further increases to 120, the fluctuation of the cost value is minimal and almost remains at the same level. After $m = 90$, further increasing m has a negligible effect on C_{avg} .

Therefore, whether increasing the capacity of umbrella rental stands may be a worthwhile consideration.

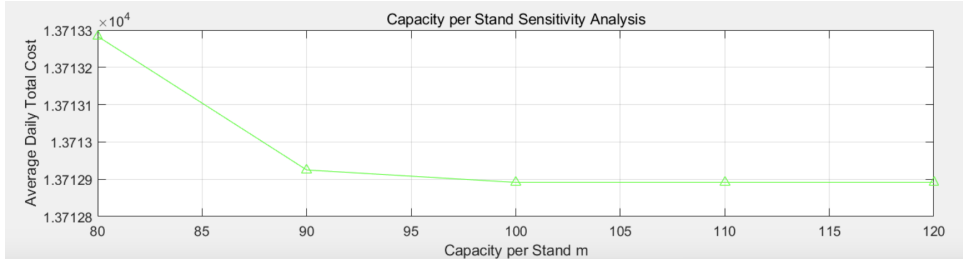


Figure 6: Capacity per Stand Sensitivity Analysis

Part V: Conclusion

This study takes ‘minimizing the cost of campus umbrella rental point layout’ as the core problem, and systematically proposes an optimal solution by building a mixed integer planning (MIP) model. Based on an in-depth analysis of the geographical features, pedestrian flow distribution and weather patterns of the campus, we first constructed a multi-objective optimization framework including construction cost, time cost and service coverage. In the modeling process, we fully consider the realistic constraints such as capacity limitations of the rental sites, user accessibility requirements and weather fluctuations to ensure the feasibility of the solution.

The study adopts MATLAB R2023b as the computational platform and uses its built-in intlinprog solver to solve the model optimally. We chose representative six days of umbrella demand data from different buildings as inputs and used parametric analysis to study the optimal layout strategy of umbrella rental stations based on this data. Among them, we conducted a comprehensive sensitivity analysis, focusing on the degree of influence of key parameters such as the construction cost coefficient, umbrella rental point cap, and capacity on the final scheme, which provides a theoretical basis for decision makers to make flexible adjustments.

The experimental results show that the proposed MIP model can effectively balance the construction investment and service benefits, and significantly reduce the total cost while ensuring a high degree of regional coverage. The optimization scheme not only gives the spatial distribution of the optimal daily rental points, but also quantifies the economic differences between different resource allocation schemes. These findings provide an important reference for the intelligent layout of campus public service facilities, and the established methodological framework can also be extended and applied to the optimal

allocation problem of other public facilities such as shared bicycle parking points and vending machines. Future research can further incorporate real-time weather forecast data and dynamic demand models to improve the adaptability and accuracy of the solution.

Part VII: Appendix

```

1 % Parameter settings
2 c = 25;           % Cost per umbrella
3 n = 10;          % Maximum number of rental stands
4 m = 100;         % Maximum capacity per stand
5 days = 6;        % Total number of days
6
7 % Building list and stand numbers
8 buildings = ␣
9   ␣ {'FB', 'CB', 'SA', 'SB', 'SC', 'SD', 'PB', 'MA', 'MB', 'EB', 'EE', 'AS', 'IR', 'IA', 'BS', 'P
10
11 % Load demand data
12 d_data = [
13 40 50 45 60 80 30;
14 60 55 35 45 50 60;
15 15 20 20 10 15 25;
16 20 25 30 15 20 25;
17 15 20 10 20 5 15;
18 20 15 18 17 16 19;
19 20 25 22 23 34 21;
20 30 32 26 50 20 15;
21 30 28 44 52 38 22;
22 25 43 21 34 26 27;
23 20 34 21 42 28 36;
24 15 11 33 23 16 17;
25 10 14 15 12 14 13;
26 30 22 33 24 31 27;
27 70 66 58 62 73 68;
28 15 14 22 17 12 24;
29 30 26 27 28 29 31;
30 30 22 35 28 34 21;
31 20 24 32 31 25 17;
32 ];
33
34 % Load time matrix
35 t_ij = [
36 0.1 0.3 0.5 0.4 0.35 0.45 0.5 0.55 0.6 0.6 0.85 0.75 0.9 0.75 0.9 0.8 ␣
37   ␣ 1 0.95 0.9;
38 0.25 0.1 0.3 0.25 0.25 0.4 0.45 0.5 0.5 0.4 0.65 0.55 0.7 0.55 0.7 ␣
39   ␣ 0.6 0.8 0.75 0.7;
40 0.3 0.25 0.2 0.1 0.05 0.2 0.25 0.3 0.4 0.35 0.6 0.5 0.65 0.5 0.65 ␣
41   ␣ 0.55 0.75 0.7 0.65;
42 0.35 0.2 0.2 0.05 0.05 0.15 0.2 0.25 0.3 0.25 0.5 0.4 0.55 0.4 0.55 ␣
43   ␣ 0.45 0.65 0.6 0.55;
44 0.4 0.25 0.1 0.1 0.15 0.2 0.25 0.3 0.25 0.15 0.4 0.3 0.45 0.3 0.45 ␣
45   ␣ 0.35 0.55 0.5 0.45;
46 0.5 0.3 0.05 0.2 0.25 0.3 0.35 0.4 0.3 0.15 0.4 0.3 0.45 0.3 0.45 ␣
47   ␣ 0.35 0.55 0.5 0.45;
48 0.4 0.35 0.3 0.2 0.15 0.05 0.1 0.15 0.3 0.25 0.5 0.4 0.55 0.4 0.55 ␣
49   ␣ 0.45 0.65 0.6 0.55;

```

```

43 0.45 0.45 0.35 0.25 0.25 0.15 0 0.05 0.2 0.25 0.5 0.4 0.55 0.4 0.55 ↵
    ↵ 0.45 0.65 0.6 0.55;
44 0.55 0.5 0.4 0.3 0.3 0.2 0.15 0.1 0.15 0.25 0.5 0.4 0.55 0.4 0.55 ↵
    ↵ 0.45 0.65 0.6 0.55;
45 0.6 0.5 0.3 0.3 0.35 0.25 0.2 0.15 0 0.15 0.4 0.3 0.45 0.3 0.45 0.35 ↵
    ↵ 0.55 0.5 0.45;
46 0.6 0.4 0.15 0.2 0.25 0.2 0.25 0.25 0.15 0.05 0.3 0.2 0.35 0.2 0.35 ↵
    ↵ 0.25 0.45 0.4 0.35;
47 0.85 0.65 0.4 0.5 0.55 0.5 0.55 0.55 0.4 0.3 0 0.1 0.2 0.25 0.4 0.2 ↵
    ↵ 0.4 0.45 0.5;
48 0.7 0.5 0.3 0.35 0.4 0.35 0.4 0.4 0.3 0.2 0.1 0 0.15 0.15 0.3 0.15 ↵
    ↵ 0.4 0.4 0.4;
49 0.7 0.5 0.3 0.35 0.4 0.35 0.4 0.4 0.3 0.2 0.25 0.15 0.25 0.05 0.2 0.1 ↵
    ↵ 0.3 0.25 0.25;
50 0.9 0.7 0.5 0.55 0.6 0.55 0.6 0.6 0.5 0.4 0.2 0.2 0.05 0.25 0.2 0.1 ↵
    ↵ 0.15 0.2 0.4;
51 0.85 0.65 0.45 0.5 0.55 0.5 0.55 0.55 0.45 0.35 0.4 0.3 0.2 0.1 0.15 ↵
    ↵ 0.1 0.15 0.1 0.3;
52 1.0 0.8 0.6 0.65 0.7 0.65 0.7 0.7 0.6 0.5 0.4 0.35 0.2 0.35 0.2 0.25 ↵
    ↵ 0.1 0.2 0.5;
53 0.95 0.75 0.55 0.6 0.65 0.6 0.65 0.65 0.55 0.45 0.45 0.35 0.3 0.25 ↵
    ↵ 0.15 0.25 0.2 0.1 0.4;
54 0.85 0.65 0.45 0.5 0.55 0.5 0.55 0.55 0.45 0.35 0.5 0.4 0.45 0.2 0.25 ↵
    ↵ 0.3 0.5 0.4 0.05;
55 ];
56
57 % Preallocate results
58 daily_costs = zeros(days, 1);
59 all_selected = cell(days, 1);
60 all_capacities = cell(days, 1);
61
62 for day = 1:days
63     fprintf('\n===== Solving Day %d =====\n', day);
64     d = d_data(:, day); % Get current day's demand
65
66     % ===== Variable Definition =====
67     num_x = 19*19; % Number of x_ij variables
68     num_y = 19; % Number of y_j variables
69     num_u = 19; % Number of u_j variables
70     total_vars = num_x + num_y + num_u;
71
72     % ===== Objective Function =====
73     f = zeros(total_vars, 1);
74     % User inconvenience cost term: sum(t_ij * d_i * x_ij)
75     for i = 1:19
76         for j = 1:19
77             idx = (i-1)*19 + j;
78             f(idx) = t_ij(i,j) * d(i);
79         end
80     end
81     % Installation cost term: c * sum(u_j)
82     f(num_x + num_y + 1 : end) = c;
83
84     % ===== Constraints Setup =====
85     % Equality constraints: Each building must be assigned to exactly ↵
        ↵ one rental stand
86     Aeq = sparse(19, total_vars);
87     beq = ones(19, 1);

```

```

88     for i = 1:19
89         cols = (i-1)*19 + (1:19);
90         Aeq(i, cols) = 1;
91     end
92
93     % Inequality constraints
94     % 1. Maximum number of rental stands:  $\sum(y_j) \leq n$ 
95     A1 = sparse(1, total_vars);
96     A1(num_x + 1 : num_x + num_y) = 1;
97     b1 = n;
98
99     % 2. Capacity constraint 1:  $\sum(d_i * x_{ij}) \leq u_j$ 
100    A2 = sparse(19, total_vars);
101    for j = 1:19
102        for i = 1:19
103            idx = (i-1)*19 + j;
104            A2(j, idx) = d(i);
105        end
106        A2(j, num_x + num_y + j) = -1; %  $-u_j$ 
107    end
108    b2 = zeros(19, 1);
109
110    % 3. Capacity constraint 2:  $u_j \leq m * y_j$ 
111    A3 = sparse(19, total_vars);
112    for j = 1:19
113        A3(j, num_x + j) = -m; %  $-m * y_j$ 
114        A3(j, num_x + num_y + j) = 1; %  $u_j$ 
115    end
116    b3 = zeros(19, 1);
117
118    % 4. Logical constraint:  $x_{ij} \leq y_j$ 
119    A4 = sparse(num_x, total_vars);
120    row = 0;
121    for i = 1:19
122        for j = 1:19
123            row = row + 1;
124            A4(row, (i-1)*19 + j) = 1; %  $x_{ij}$ 
125            A4(row, num_x + j) = -1; %  $-y_j$ 
126        end
127    end
128    b4 = zeros(num_x, 1);
129
130    % Combine all constraints
131    A = [A1; A2; A3; A4];
132    b = [b1; b2; b3; b4];
133
134    % ===== Variable Bounds =====
135    lb = zeros(total_vars, 1);
136    ub = ones(total_vars, 1);
137    ub(num_x + num_y + 1 : end) = m; %  $u_j \leq m$ 
138
139    % ===== Integer Constraints =====
140    intcon = 1:(num_x + num_y); %  $x_{ij}$  and  $y_j$  are binary
141    intcon = [intcon, num_x+num_y+1:total_vars]; %  $u_j$  are integers
142
143    % ===== Solve MIP =====
144    opts = optimoptions('intlinprog', 'Display', 'final', ↵
        ↵ 'Heuristics', 'advanced');

```

```

145 [sol, total_cost, exitflag] = intlinprog(f, intcon, A, b, Aeq, ←
    ↪ beq, lb, ub, opts);
146
147 if exitflag ~= 1
148     error('Failed to solve Day%d, Exit flag: %d', day, exitflag);
149 end
150
151 % ===== Result Parsing =====
152 y = sol(num_x + 1 : num_x + num_y) > 0.5;
153 u = round(sol(num_x + num_y + 1 : end));
154
155 % Save results
156 daily_costs(day) = total_cost;
157 all_selected{day} = find(y);
158 all_capacities{day} = u(y);
159
160 % ===== Result Output =====
161 fprintf(' Day %d Optimization Results \n', day);
162 fprintf('Total Cost: %.2f\n', total_cost);
163 fprintf('Enabled Rental Stands: ');
164 fprintf('%d ', all_selected{day});
165 fprintf('\nCapacity Configuration for Each Stand:\n');
166 for k = 1:length(all_selected{day})
167     j = all_selected{day}(k);
168     fprintf(' Stand %2d: %3d umbrellas\n', j, ←
    ↪ all_capacities{day}(k));
169 end
170 end
171
172 % ===== Final Report =====
173 fprintf('\n==== Final Optimization Report ==== \n');
174 for day = 1:days
175     fprintf('Day%d: Total Cost=%.2f, Number of Stands Enabled=%d\n',...
176         day, daily_costs(day), length(all_selected{day}));
177 end
178
179
180
181 %% ===== Sensitivity Analysis =====
182 % Define parameter ranges for analysis
183 c_values = 20:2:30; % Installation cost analysis
184 n_values = 5:15; % Maximum rental stands analysis
185 m_values = 80:10:120; % Capacity per stand analysis
186
187 % Preallocate results storage
188 %% Installation Cost Sensitivity Analysis
189 fprintf('\n==== Starting Installation Cost Sensitivity Analysis ←
    ↪ ==== \n');
190 for c_idx = 1:length(c_values)
191     current_c = c_values(c_idx);
192     day_costs = zeros(days,1);
193     day_stands = zeros(days,1);
194
195     for day = 1:days
196         % Get current day's demand
197         d = d_data(:, day);
198
199         % Construct objective function

```

```

200     num_x = 19*19;
201     num_y = 19;
202     num_u = 19;
203     total_vars = num_x + num_y + num_u;
204
205     f = zeros(total_vars, 1);
206     for i = 1:19
207         for j = 1:19
208             idx = (i-1)*19 + j;
209             f(idx) = t_ij(i,j) * d(i);
210         end
211     end
212     f(num_x + num_y + 1 : end) = current_c;
213
214     % Construct constraints
215     % Equality constraints
216     Aeq = sparse(19, total_vars);
217     beq = ones(19, 1);
218     for i = 1:19
219         cols = (i-1)*19 + (1:19);
220         Aeq(i, cols) = 1;
221     end
222
223     % Inequality constraints
224     A1 = sparse(1, total_vars);
225     A1(num_x + 1 : num_x + num_y) = 1;
226     b1 = n;
227
228     A2 = sparse(19, total_vars);
229     for j = 1:19
230         for i = 1:19
231             idx = (i-1)*19 + j;
232             A2(j, idx) = d(i);
233         end
234         A2(j, num_x + num_y + j) = -1;
235     end
236     b2 = zeros(19, 1);
237
238     A3 = sparse(19, total_vars);
239     for j = 1:19
240         A3(j, num_x + j) = -m;
241         A3(j, num_x + num_y + j) = 1;
242     end
243     b3 = zeros(19, 1);
244
245     A4 = sparse(num_x, total_vars);
246     row = 0;
247     for i = 1:19
248         for j = 1:19
249             row = row + 1;
250             A4(row, (i-1)*19 + j) = 1;
251             A4(row, num_x + j) = -1;
252         end
253     end
254     b4 = zeros(num_x, 1);
255
256     A = [A1; A2; A3; A4];
257     b = [b1; b2; b3; b4];

```

```

258
259 % Variable bounds
260 lb = zeros(total_vars, 1);
261 ub = ones(total_vars, 1);
262 ub(num_x + num_y + 1 : end) = m;
263
264 % Integer constraints
265 intcon = 1:(num_x + num_y);
266 intcon = [intcon, num_x+num_y+1:total_vars];
267
268 % Solve
269 opts = optimoptions('intlinprog', 'Display', 'none');
270 [sol, total_cost, exitflag] = intlinprog(f, intcon, A, b, ←
    ↪ Aeq, beq, lb, ub, opts);
271
272 if exitflag ~= 1
273     warning('Failed to solve Day%d, Exit flag: %d', day, ←
    ↪ exitflag);
274     day_costs(day) = Inf;
275     day_stands(day) = Inf;
276 else
277     % Record results
278     day_costs(day) = total_cost;
279     y = sol(num_x + 1 : num_x + num_y) > 0.5;
280     day_stands(day) = sum(y);
281 end
282 end
283
284 sensitivity_results.c.costs(c_idx,:) = mean(day_costs);
285 sensitivity_results.c.stands(c_idx,:) = mean(day_stands);
286 end
287
288 %% Maximum Rental Stands Sensitivity Analysis
289 fprintf('\n==== Starting Maximum Rental Stands Sensitivity Analysis ←
    ↪ ==== \n');
290 for n_idx = 1:length(n_values)
291     current_n = n_values(n_idx);
292     day_costs = zeros(days,1);
293     day_stands = zeros(days,1);
294
295     for day = 1:days
296         % Get current day's demand
297         d = d_data(:, day);
298
299         % Construct objective function
300         num_x = 19*19;
301         num_y = 19;
302         num_u = 19;
303         total_vars = num_x + num_y + num_u;
304
305         f = zeros(total_vars, 1);
306         for i = 1:19
307             for j = 1:19
308                 idx = (i-1)*19 + j;
309                 f(idx) = t_ij(i,j) * d(i);
310             end
311         end
312         f(num_x + num_y + 1 : end) = c;

```

```

313
314 % Construct constraints
315 % Equality constraints
316 Aeq = sparse(19, total_vars);
317 beq = ones(19, 1);
318 for i = 1:19
319     cols = (i-1)*19 + (1:19);
320     Aeq(i, cols) = 1;
321 end
322
323 % Inequality constraints
324 A1 = sparse(1, total_vars);
325 A1(num_x + 1 : num_x + num_y) = 1;
326 b1 = current_n;
327
328 A2 = sparse(19, total_vars);
329 for j = 1:19
330     for i = 1:19
331         idx = (i-1)*19 + j;
332         A2(j, idx) = d(i);
333     end
334     A2(j, num_x + num_y + j) = -1;
335 end
336 b2 = zeros(19, 1);
337
338 A3 = sparse(19, total_vars);
339 for j = 1:19
340     A3(j, num_x + j) = -m;
341     A3(j, num_x + num_y + j) = 1;
342 end
343 b3 = zeros(19, 1);
344
345 A4 = sparse(num_x, total_vars);
346 row = 0;
347 for i = 1:19
348     for j = 1:19
349         row = row + 1;
350         A4(row, (i-1)*19 + j) = 1;
351         A4(row, num_x + j) = -1;
352     end
353 end
354 b4 = zeros(num_x, 1);
355
356 A = [A1; A2; A3; A4];
357 b = [b1; b2; b3; b4];
358
359 % Variable bounds
360 lb = zeros(total_vars, 1);
361 ub = ones(total_vars, 1);
362 ub(num_x + num_y + 1 : end) = m;
363
364 % Integer constraints
365 intcon = 1:(num_x + num_y);
366 intcon = [intcon, num_x+num_y+1:total_vars];
367
368 % Solve
369 opts = optimoptions('intlinprog', 'Display', 'none');

```

```

370     [sol, total_cost, exitflag] = intlinprog(f, intcon, A, b, ←
        ↪ Aeq, beq, lb, ub, opts);
371
372     if exitflag ~= 1
373         warning('Failed to solve Day%d, Exit flag: %d', day, ←
            ↪ exitflag);
374         day_costs(day) = Inf;
375         day_stands(day) = Inf;
376     else
377         % Record results
378         day_costs(day) = total_cost;
379         y = sol(num_x + 1 : num_x + num_y) > 0.5;
380         day_stands(day) = sum(y);
381     end
382 end
383
384 sensitivity_results.n.costs(n_idx,:) = mean(day_costs);
385 sensitivity_results.n.stands(n_idx,:) = mean(day_stands);
386 end
387
388 %% Capacity per Stand Sensitivity Analysis
389 fprintf('\n==== Starting Capacity per Stand Sensitivity Analysis ←
    ↪ =====\n');
390 for m_idx = 1:length(m_values)
391     current_m = m_values(m_idx);
392     day_costs = zeros(days,1);
393     day_stands = zeros(days,1);
394
395     for day = 1:days
396         % Get current day's demand
397         d = d_data(:, day);
398
399         % Construct objective function
400         num_x = 19*19;
401         num_y = 19;
402         num_u = 19;
403         total_vars = num_x + num_y + num_u;
404
405         f = zeros(total_vars, 1);
406         for i = 1:19
407             for j = 1:19
408                 idx = (i-1)*19 + j;
409                 f(idx) = t_ij(i,j) * d(i);
410             end
411         end
412         f(num_x + num_y + 1 : end) = c;
413
414         % Construct constraints
415         % Equality constraints
416         Aeq = sparse(19, total_vars);
417         beq = ones(19, 1);
418         for i = 1:19
419             cols = (i-1)*19 + (1:19);
420             Aeq(i, cols) = 1;
421         end
422
423         % Inequality constraints
424         A1 = sparse(1, total_vars);

```

```

425     A1(num_x + 1 : num_x + num_y) = 1;
426     b1 = n;
427
428     A2 = sparse(19, total_vars);
429     for j = 1:19
430         for i = 1:19
431             idx = (i-1)*19 + j;
432             A2(j, idx) = d(i);
433         end
434         A2(j, num_x + num_y + j) = -1;
435     end
436     b2 = zeros(19, 1);
437
438     A3 = sparse(19, total_vars);
439     for j = 1:19
440         A3(j, num_x + j) = -current_m;
441         A3(j, num_x + num_y + j) = 1;
442     end
443     b3 = zeros(19, 1);
444
445     A4 = sparse(num_x, total_vars);
446     row = 0;
447     for i = 1:19
448         for j = 1:19
449             row = row + 1;
450             A4(row, (i-1)*19 + j) = 1;
451             A4(row, num_x + j) = -1;
452         end
453     end
454     b4 = zeros(num_x, 1);
455
456     A = [A1; A2; A3; A4];
457     b = [b1; b2; b3; b4];
458
459     % Variable bounds
460     lb = zeros(total_vars, 1);
461     ub = ones(total_vars, 1);
462     ub(num_x + num_y + 1 : end) = current_m;
463
464     % Integer constraints
465     intcon = 1:(num_x + num_y);
466     intcon = [intcon, num_x+num_y+1:total_vars];
467
468     % Solve
469     opts = optimoptions('intlinprog', 'Display', 'none');
470     [sol, total_cost, exitflag] = intlinprog(f, intcon, A, b, ←
        ↪ Aeq, beq, lb, ub, opts);
471
472     if exitflag ~= 1
473         warning('Failed to solve Day%d, Exit flag: %d', day, ←
            ↪ exitflag);
474         day_costs(day) = Inf;
475         day_stands(day) = Inf;
476     else
477         % Record results
478         day_costs(day) = total_cost;
479         y = sol(num_x + 1 : num_x + num_y) > 0.5;
480         day_stands(day) = sum(y);

```

```

481         end
482     end
483
484     sensitivity_results.m.costs(m_idx,:) = mean(day_costs);
485     sensitivity_results.m.stands(m_idx,:) = mean(day_stands);
486 end
487
488 %% ===== Visualization of Results =====
489 figure('Position',[100 100 1000 800])
490
491 % Installation Cost Analysis
492 subplot(3,1,1)
493 plot(sensitivity_results.c.values, sensitivity_results.c.costs, 'b-o')
494 title('Installation Cost Sensitivity Analysis')
495 xlabel('Installation Cost c'), ylabel('Average Daily Total Cost')
496 grid on
497
498 subplot(3,1,2)
499 plot(sensitivity_results.n.values, sensitivity_results.n.costs, 'r-s')
500 title('Maximum Rental Stands Sensitivity Analysis')
501 xlabel('Maximum Number of Rental Stands n'), ylabel('Average Daily ↵
    ↵ Total Cost')
502 grid on
503
504 subplot(3,1,3)
505 plot(sensitivity_results.m.values, sensitivity_results.m.costs, 'g-^')
506 title('Capacity per Stand Sensitivity Analysis')
507 xlabel('Capacity per Stand m'), ylabel('Average Daily Total Cost')
508 grid on
509
510
511
512
513
514
515 %% The following code cannot be visualized
516
517 %% ===== Demand Sensitivity Analysis =====
518 % Define parameter ranges for analysis
519 d_scale = 0.5:0.1:1.5; % Demand scaling factors
520
521 % Preallocate results storage
522 demand_results = struct('values', d_scale, 'costs', [], 'stands', []);
523
524 %% Demand Sensitivity Analysis
525 fprintf('\n===== Starting Demand Sensitivity Analysis =====\n');
526 for scale_idx = 1:length(d_scale)
527     current_scale = d_scale(scale_idx);
528     day_costs = zeros(days,1);
529     day_stands = zeros(days,1);
530
531     for day = 1:days
532         % Get current day's demand and scale it
533         d = d_data(:, day) * current_scale;
534
535         % Construct objective function
536         num_x = 19*19;
537         num_y = 19;

```

```

538     num_u = 19;
539     total_vars = num_x + num_y + num_u;
540
541     f = zeros(total_vars, 1);
542     for i = 1:19
543         for j = 1:19
544             idx = (i-1)*19 + j;
545             f(idx) = t_ij(i,j) * d(i);
546         end
547     end
548     f(num_x + num_y + 1 : end) = c;
549
550     % Construct constraints
551     % Equality constraints
552     Aeq = sparse(19, total_vars);
553     beq = ones(19, 1);
554     for i = 1:19
555         cols = (i-1)*19 + (1:19);
556         Aeq(i, cols) = 1;
557     end
558
559     % Inequality constraints
560     A1 = sparse(1, total_vars);
561     A1(num_x + 1 : num_x + num_y) = 1;
562     b1 = n;
563
564     A2 = sparse(19, total_vars);
565     for j = 1:19
566         for i = 1:19
567             idx = (i-1)*19 + j;
568             A2(j, idx) = d(i);
569         end
570         A2(j, num_x + num_y + j) = -1;
571     end
572     b2 = zeros(19, 1);
573
574     A3 = sparse(19, total_vars);
575     for j = 1:19
576         A3(j, num_x + j) = -m;
577         A3(j, num_x + num_y + j) = 1;
578     end
579     b3 = zeros(19, 1);
580
581     A4 = sparse(num_x, total_vars);
582     row = 0;
583     for i = 1:19
584         for j = 1:19
585             row = row + 1;
586             A4(row, (i-1)*19 + j) = 1;
587             A4(row, num_x + j) = -1;
588         end
589     end
590     b4 = zeros(num_x, 1);
591
592     A = [A1; A2; A3; A4];
593     b = [b1; b2; b3; b4];
594
595     % Variable bounds

```

```

596     lb = zeros(total_vars, 1);
597     ub = ones(total_vars, 1);
598     ub(num_x + num_y + 1 : end) = m;
599
600     % Integer constraints
601     intcon = 1:(num_x + num_y);
602     intcon = [intcon, num_x+num_y+1:total_vars];
603
604     % Solve
605     opts = optimoptions('intlinprog', 'Display', 'none');
606     [sol, total_cost, exitflag] = intlinprog(f, intcon, A, b, ←
        ↪ Aeq, beq, lb, ub, opts);
607
608     if exitflag ~= 1
609         warning('Failed to solve Day%d, Exit flag: %d', day, ←
            ↪ exitflag);
610         day_costs(day) = Inf;
611         day_stands(day) = Inf;
612     else
613         % Record results
614         day_costs(day) = total_cost;
615         y = sol(num_x + 1 : num_x + num_y) > 0.5;
616         day_stands(day) = sum(y);
617     end
618 end
619
620 % Save results
621 demand_results.costs(scale_idx) = mean(day_costs);
622 demand_results.stands(scale_idx) = mean(day_stands);
623 end
624
625 %% ===== Visualization of Demand Sensitivity Analysis ←
    ↪ Results =====
626 figure('Position',[100 100 1000 400])
627
628 % Demand Sensitivity Analysis
629 plot(demand_results.values, [demand_results.costs], 'k-d')
630 title('Demand Sensitivity Analysis')
631 xlabel('Demand Scaling Factor'), ylabel('Average Daily Total Cost')
632 grid on

```

```

===== Solving Day 1 =====

找到最优解。

Intlinprog 在根节点处停止, 因为目标值在最优值的间隙容差范围内, options.AbsoluteGapTolerance = 0. intcon 变量是
容差范围内的整数, options.IntegerTolerance = 1e-05.

【Day1 Optimization Results】
Total Cost: 12919.25
Enabled Rental Stands: 1 2 3 5 7 9 11 13 14 18
Capacity Configuration for Each Stand:
Stand 1: 40 umbrellas
Stand 2: 60 umbrellas
Stand 3: 35 umbrellas
Stand 5: 35 umbrellas
Stand 7: 50 umbrellas
Stand 9: 75 umbrellas
Stand 11: 25 umbrellas
Stand 13: 70 umbrellas
Stand 14: 50 umbrellas
Stand 18: 75 umbrellas

===== Solving Day 2 =====

找到最优解。

Intlinprog 在根节点处停止, 因为目标值在最优值的间隙容差范围内, options.AbsoluteGapTolerance = 0. intcon 变量是
容差范围内的整数, options.IntegerTolerance = 1e-05.

【Day2 Optimization Results】
Total Cost: 13694.10
Enabled Rental Stands: 1 2 5 7 9 10 12 13 18 19
Capacity Configuration for Each Stand:
Stand 1: 50 umbrellas
Stand 2: 55 umbrellas
Stand 5: 45 umbrellas
Stand 7: 57 umbrellas
Stand 9: 71 umbrellas
Stand 10: 69 umbrellas
Stand 12: 47 umbrellas
Stand 13: 66 umbrellas
Stand 18: 62 umbrellas
Stand 19: 24 umbrellas

===== Solving Day 3 =====

找到最优解。

Intlinprog 在根节点处停止, 因为目标值在最优值的间隙容差范围内, options.AbsoluteGapTolerance = 0. intcon 变量是
容差范围内的整数, options.IntegerTolerance = 1e-05.

【Day3 Optimization Results】
Total Cost: 13721.45
Enabled Rental Stands: 1 2 3 5 8 11 13 14 18 19
Capacity Configuration for Each Stand:
Stand 1: 45 umbrellas
Stand 2: 35 umbrellas
Stand 3: 49 umbrellas
Stand 5: 72 umbrellas
Stand 8: 91 umbrellas
Stand 11: 48 umbrellas
Stand 13: 58 umbrellas
Stand 14: 33 umbrellas
Stand 18: 84 umbrellas
Stand 19: 32 umbrellas

===== Solving Day 4 =====

找到最优解。

Intlinprog 在根节点处停止, 因为目标值在最优值的间隙容差范围内, options.AbsoluteGapTolerance = 0. intcon 变量是
容差范围内的整数, options.IntegerTolerance = 1e-05.

【Day4 Optimization Results】
Total Cost: 14874.65
Enabled Rental Stands: 1 2 4 7 9 10 12 13 18 19
Capacity Configuration for Each Stand:
Stand 1: 60 umbrellas
Stand 2: 45 umbrellas
Stand 4: 45 umbrellas
Stand 7: 73 umbrellas
Stand 9: 86 umbrellas
Stand 10: 59 umbrellas
Stand 12: 59 umbrellas
Stand 13: 62 umbrellas
Stand 18: 73 umbrellas
Stand 19: 31 umbrellas

```

Figure 7:

```

===== Solving Day 5 =====

找到最优解。

Intlinprog 在根节点处停止，因为目标值在最优值的间隙容差范围内，options.AbsoluteGapTolerance = 0. intcon 变量是
容差范围内的整数，options.IntegerTolerance = 1e-05。

【Day5 Optimization Results】
Total Cost: 14199.95
Enabled Rental Stands: 1 2 5 7 9 10 12 13 18 19
Capacity Configuration for Each Stand:
  Stand 1: 80 umbrellas
  Stand 2: 50 umbrellas
  Stand 5: 35 umbrellas
  Stand 7: 92 umbrellas
  Stand 9: 26 umbrellas
  Stand 10: 49 umbrellas
  Stand 12: 61 umbrellas
  Stand 13: 73 umbrellas
  Stand 18: 75 umbrellas
  Stand 19: 25 umbrellas

===== Solving Day 6 =====

找到最优解。

Intlinprog 在根节点处停止，因为目标值在最优值的间隙容差范围内，options.AbsoluteGapTolerance = 0. intcon 变量是
容差范围内的整数，options.IntegerTolerance = 1e-05。

【Day6 Optimization Results】
Total Cost: 12867.95
Enabled Rental Stands: 1 2 5 7 9 10 11 13 14 17
Capacity Configuration for Each Stand:
  Stand 1: 30 umbrellas
  Stand 2: 60 umbrellas
  Stand 5: 65 umbrellas
  Stand 7: 58 umbrellas
  Stand 9: 27 umbrellas
  Stand 10: 55 umbrellas
  Stand 11: 30 umbrellas
  Stand 13: 68 umbrellas
  Stand 14: 68 umbrellas
  Stand 17: 52 umbrellas

===== Final Optimization Report =====
Day1: Total Cost=12919.25, Number of Stands Enabled=10
Day2: Total Cost=13694.10, Number of Stands Enabled=10
Day3: Total Cost=13721.45, Number of Stands Enabled=10
Day4: Total Cost=14874.65, Number of Stands Enabled=10
Day5: Total Cost=14199.95, Number of Stands Enabled=10
Day6: Total Cost=12867.95, Number of Stands Enabled=10

===== Starting Installation Cost Sensitivity Analysis =====

===== Starting Maximum Rental Stands Sensitivity Analysis =====
警告: Failed to solve Day1, Exit flag: -2
> 位置: MIP2 (第 373 行)
警告: Failed to solve Day2, Exit flag: -2
> 位置: MIP2 (第 373 行)
警告: Failed to solve Day3, Exit flag: -2
> 位置: MIP2 (第 373 行)
警告: Failed to solve Day4, Exit flag: -2
> 位置: MIP2 (第 373 行)
警告: Failed to solve Day5, Exit flag: -2
> 位置: MIP2 (第 373 行)
警告: Failed to solve Day6, Exit flag: -2
> 位置: MIP2 (第 373 行)

===== Starting Capacity per Stand Sensitivity Analysis =====

===== Starting Demand Sensitivity Analysis =====
警告: Failed to solve Day1, Exit flag: -1
> 位置: MIP2 (第 609 行)
警告: Failed to solve Day3, Exit flag: -1
> 位置: MIP2 (第 609 行)

>> results
数据已成功写入到 Optimization_Results.xlsx 文件中。

```

Figure 8: